

TED[15] 6021

Reg. No.....

(REVISION-2015)

Signature.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE
MODEL QUESTION PAPER

ADVANCED PRODUCTION PROCESSES

[Time: 3 hours]

(Maximum marks: 100)

PART - A

(Maximum marks: 10)

Marks

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define the term automation.
2. State the advantages of broaching.
3. Differentiate truing and dressing of grinding wheel.
4. Describe the basic elements of a robot.
5. List two methods of gear manufacture.

(5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain the various tool holding used in capstan lathe and turret lathe.
2. Illustrate a pull type broach tool with necessary sketch.
3. Distinguish honing and lapping process?
4. Describe about AGVS.
5. Explain the hydraulic servo copying system with the help of a neat sketch.
6. Describe the principle of gear shaping.
7. Explain the principle of laser beam machining with a neat sketch.

(5x6=30)

PART - C
(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

- III.a. Explain the parts of a turret lathe with a neat sketch. (8)
- b. Describe pantograph copying system with the help of necessary sketch (7)

OR

- IV.a. Explain the turret head indexing mechanism with a neat sketch (8)
- b. Illustrate the electric copying system with a neat sketch (7)

UNIT – II

- V.a. Explain the classification of dies used in press working. (8)
- b. Differentiate jigs and fixtures. (7)

OR

- VI.a. Illustrate the working of any two drill jigs with the help of neat sketch (8)
- b. Describe the gear hobbing process with a neat sketch (7)

UNIT – III

- VII.a. Describe the working of centre less grinder with suitable figure (8)
- b. Explain the working of USM with a neat sketch (7)

OR

- VIII.a. Explain electro chemical machining with the help of neat sketch (8)
- b. Describe the following surface finishing methods : a) Anodising b) Metal spraying (7)

UNIT – IV

- IX.a. Explain flexible manufacturing cell with the help of block diagram (8)
- b. Discuss about various applications of robots. (7)

OR

- X.a. Explain the various components of a NC machine with block diagram (8)
- b. Mention the advantages of using CAD and CAM (7)